

A Bluetooth Low-Energy (BLE) Transceiver with TX/RX Switchable On-Chip Matching Network, 2.75mW High-IF Discrete-Time Receiver, and 3.6mW All-Digital Transmitter

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Abstract

We present a new ultra-low-power (ULP) transceiver for Internet-of-Things (IoT) optimized for 28-nm CMOS. The receiver (RX) employs a high-rate (up to 10 GS/s) discrete-time (DT) architecture with intermediate frequency (IF) placed beyond the 1/f noise corner of MOS devices. New multi-stage multi-rate charge-sharing bandpass filters are adapted to achieve high out-of-band linearity, low noise and low power consumption. A transmitter (TX) employs an all-digital PLL (ADPLL) with switched-current-source digitally controlled oscillator (DCO) and switching PA. An integrated on-chip matching network serves both PA and LNTA, thus allowing a 1-pin direct antenna connection with no external antenna filters. The transceiver consumes 2.75 mW in RX and 3.6 mW in TX when delivering 0 dBm in Bluetooth LE.

Introduction

Ultra-low power (ULP) transceivers are key subsystems for wireless sensor networks and Internet-of-Things (IoT), which impose stringent requirements on cost (i.e., low silicon area and few or no external components) and power consumption. Recent ULP receivers (RX) achieve significant power reduction [1]-[4] typically using sliding intermediate frequency (IF) and low-IF continuous-time (CT) architectures, but with a sacrificed performance. To further decrease power consumption while maintaining adequate performance, we propose a fully discrete-time (DT) high-IF RX architecture with *complex-signaling* bandpass filters (BPF) and a progressively reduced sampling rate. The new approach exploits a DT filtering recently introduced for high-performance 4G RX [5], which is adapted here for ULP.

To eliminate the bulky and costly antenna filter, two techniques are employed. First, out-of-band (OOB) blockers are attenuated using a combination of a charge sampling mixer and a full-rate ($\sim 4 \times 2.45$ GS/s) charge-sharing (CS) BPF. Second, a dynamically switched fully integrated TX/RX matching network is proposed. The TX architecture is an improvement of [6] and consists of an ADPLL with switched-current-source digitally controlled oscillator (DCO), which maximally reduces its frequency pushing and 1/f phase noise such that it can operate with only a few kHz drift in *open-loop* in RX and during TX modulation. The digital PA (DPA) operates in class-E/F₂, at V_{DD}=0.5V supply. Fig. 1 shows the complete BLE transceiver.

Discrete-time Receiver with Direct Antenna Connection

The high-IF RX implements a multi-rate DT using three stages of cascaded quadrature charge-sharing (CS) BPFs. The circuitry after the low-noise transconductance amplifier (LNTA) comprises only gm stages, switches and capacitors, which are amenable to deep CMOS scaling. Gain is provided by LNTA and 3 highly linear programmable inverter-based gm-cells with 12 dB of maximum gain each. RX frontend is detailed in Fig. 2. The narrow-band LNTA with Q-factor ~ 10 is followed by a single-to-differential passive mixer and a 4/4 CS-BPF with 25%

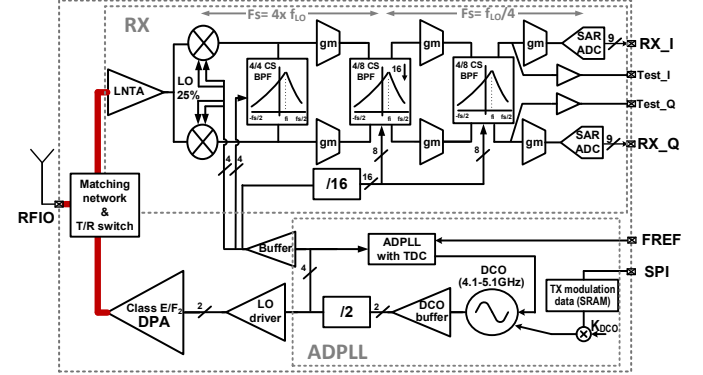


Fig. 1. Block diagram of the proposed Bluetooth LE transceiver.

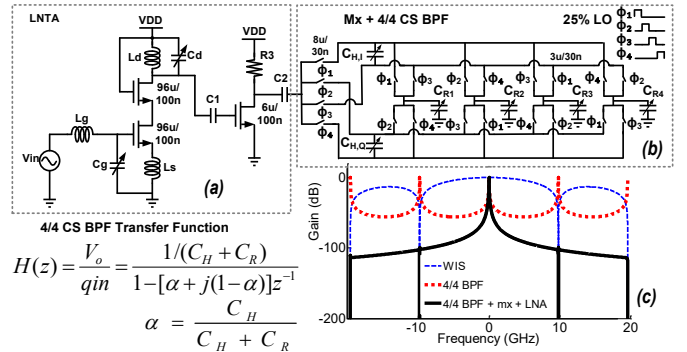


Fig. 2. LNTA and 1st-stage DT complex BPF using I/Q charge sharing.

duty-cycle clocks [5]. The mixer's charge-sampling operation at $4f_{LO}$ creates a window integration sampling (WIS) filter. Combined transfer functions of LNTA, mixer, and filter are shown in Fig. 2c and exhibit protection from OOB blockers (>100 dB at $f_s = 4f_{LO}$). The three IF BPFs provide enough anti-alias filtering for the following 9-bit SAR ADCs, which consumes only 0.125 mW each for 20 MS/s.

The 5 MHz IF processing continues with two more stages employing a new 8-phase CS-BPF structure. The 4/8 CS-BPF has 8 rotating capacitors, sampled at 8 phases with 12.5% duty-cycle (Fig. 3) and works at 16x slower clock to further reduce power consumption. The 8-phase implementation improves selectivity, with an additional 5 dB of image rejection (Fig. 3b). Decimation is performed by reducing clock frequency and integrating the samples over the rotating capacitors, creating a sinc-like effect that works as an anti-aliasing filter for the decimated filters. The

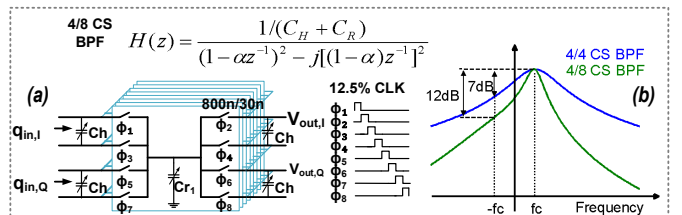


Fig. 3. 4/8 CS BPF (2nd and 3rd stages) (a); and comparison (b).

transfer function of the DT filters is accurate and controlled by the clock frequency and precisely programmable capacitor ratios to account for process, voltage and temperature (PVT) variations.

All-Digital TX and On-Chip Matching Network

The all-digital TX presented in Fig.1 consists of a two-staggered-chains TDC topology, adopted here to double the TDC resolution to 12 ps. The V_{DD} of 0.5 V was chosen for the last PA stage to deliver low output power and to abide by the process reliability rules of gate-oxide breakdown. Furthermore, class-E/ F_2 tuning exhibits the lowest systematic drain current and thus P_{out} at the same V_{DD} and load resistance among different flavors of switched-mode PAs, without sacrificing the power added efficiency (PAE). Consequently, this PA needs smaller impedance transformation ratio for $P_{out} < 3$ dBm, which results in a lower insertion loss for its matching network and thus higher system efficiency.

Fig.4a shows the proposed implementation of the on-chip matching network with a 'soft' switch between the TX and RX paths (i.e., "T/R switch"). PA's transformer-based matching network (TXMN) acts as a second-order resonator in the RX mode. The input matching of LNTA is quite sensitive to the imaginary part of the impedance Z_{TX} seen from the output pad towards TX. Hence, the resonance of PA's matching network should be adjusted via C_1 and C_2 for the operating band. Furthermore, the equivalent Q -factor and thus the input parallel resistance, R_{TX} , of TXMN should be as high as possible to exhibit the lowest penalty on the RX NF. R_{TX} is about 250Ω with less than 1 dB NF penalty in this design. In the TX mode, however, the resonant frequency of LNTA's matching network is pushed to higher frequencies via C_{gs} . Consequently, PA sees the RX path as a small capacitor in parallel with $1.2k\Omega$ modeling LNA matching network losses (see Fig.4d). Compared to the 50Ω load, they create a large impedance path for the TX signal, which leads to a negligible penalty ($\sim 4\%$) in the drain efficiency.

Measurement Results and Conclusion

Fig.4 shows measured S_{11} during LNTA and PA operations. The measured DT IF transfer function is presented in Fig.5. The desired asymmetry between positive and negative frequency offsets with respect to LO is due to the presence of complex filters. The RX gain is 46 dB with a minimum of 24 dB image rejection obtained in the IF strip, thus allowing for the budget of 31 dB to be easily achieved when combined with a typical digital baseband. OOB blocking performance shows a large margin to the BLE specification, verifying the feasibility of the direct (i.e., filter-less) antenna connection. Sensitivity of -95 dBm is measured through packet error rate (PER) curve for PER = 30.8% [2]. RX core consumes 2.75 mW with a maximum gain of 46 dB, NF of 6.5 dB and peak IIP3 of -19 dB, as shown in Fig. 5. The TX consumes 3.6 mW at 0 dBm RF output. During TX and RX packets, the ADPLL is shut down immediately after settling [6]. The DCO tuning word is maintained on its update port while the second port is used to perform an open-loop modulation (see Fig.1). This reduces the LO power from 1.4 mW to 0.6 mW. This is possible due to the very low frequency pushing (10–12 kHz/mV) and $1/f^3$ corner frequency (<100 kHz) of the DCO. BLE micrograph shows an active area of only 1.4 mm^2 in TSMC 28 nm CMOS. Power breakdown is presented in Fig. 5.

Table I summarizes the proposed transceiver and compares it with state-of-the-art BLE designs. Besides being fully integrated with an on-chip T/R switch, as in [2] and [3], this work reaches similar (NF, linearity and sensitivity) or better (max P_{out} , PLL PN) RF performance but at a much lower power consumption, even better than [1] [4], which use an off-chip matching network and T/R switch.

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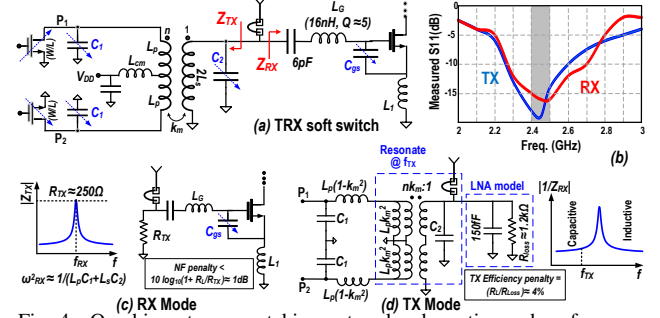


Fig. 4. On-chip antenna matching network schematics and performance.

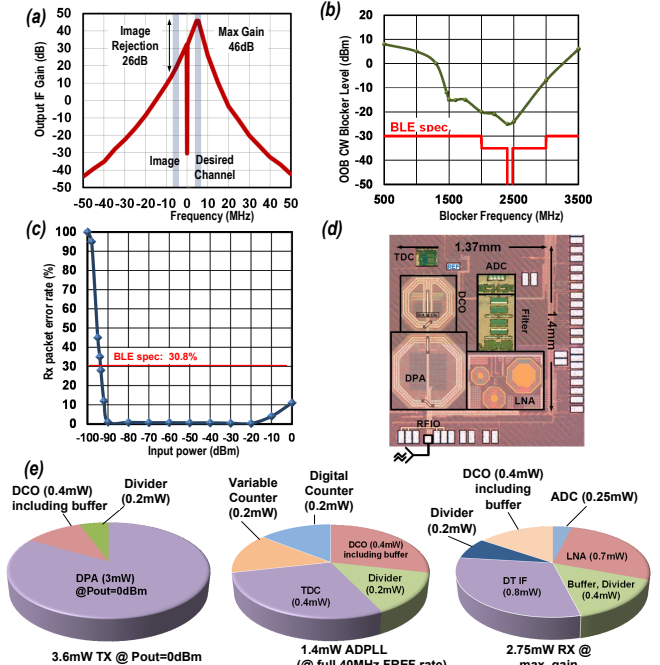


Fig. 5. Summary of chip measurements: gain, blocking test and PER measurement, power breakdown of main blocks.

TABLE I
PERFORMANCE SUMMARY AND COMPARISON WITH STATE-OF-THE-ART.

	This work	[1]	[2]	[3]	[4]	
CMOS node	28nm	90nm	55nm	40nm	40nm	
Data rate & modulation	1-Mbps GFSK	1-Mbps GFSK	1-Mbps GFSK	1-Mbps GFSK	1-Mbps GFSK	
RX noise figure (dB)	6.5	6	N/A	6.5	N/A	
RX sensitivity (dBm)	-95	-98	-94.5	-94.5	-94	
RX IIP3 (dBm)	-19	-19	N/A	N/A	N/A	
OSC FoM (dB)	188	183	N/A	N/A	N/A	
PLL in-band PN (dBc/Hz)	-92**	-85	N/A	N/A	N/A	
	-101***		N/A	N/A	N/A	
Integrated PN (degree)	1.08**	2.3	N/A	N/A	N/A	
	0.87***		N/A	N/A	N/A	
TX max. Pout (dBm)	3	0	2.3	0	-2	
Maximum PA PAE	41%	N/A	45%	<25%	N/A	
T/R Switch	Yes	No	Yes	Yes	No	
Supply voltage (V)	0.5* / 1	1.2	0.9-3.3	1.1	1	
Power consumption (mW)	TX @ 0dBm	3.6	5.4	10.1	7.7	4.2*
	RX	2.75	3.3	11.2	6.3	3.3
TX efficiency (P _{OUT} /P _{DC})		28%	18%	10%	13%	15%

*Measured @-2dBm **FREF=5MHz ***FREF=40MHz *Low-voltage DCO & DPA